Backend Home Assignment — Expression Evaluation Engine

The task: Build a library that evaluates expressions like 2 + 3 * (x - 1) against a provided set of named variables. Picture our product letting data scientists define computed variables and logics using these expressions. Feel more than free to use claude and/or other ai tools for this assignment.

Context — please read carefully:

This engine is going to run millions of times per second in production, usually for similar expressions.

We plan to open-source it once it's mature.

Other backend teams in the company will pull it as a dependency.

Make sure we support booleans and conditional operators like: ternary if,and, or, etc..

Make sure expressions can also handle strings.

Allow to define and use variables inside expressions.

Support +, -, *, /, parentheses, and variable lookup. We'd also like common math operations available.

Support defining and using functions inside expressions.

It should be thread-safe

Null and undefined values are handled throughout evaluation.

What to submit:

Working code. Choose from either python or go.

A short write-up (max 1 page.(

How did you approach this? What did you decide vs. what did the AI decide? Some instructions are vague by design, tell us what you decided and why.

Describe the library. How to use it, and what kind of features and expressions it supports.

Your AI chat transcripts. Whatever you used. If your tool stores them locally (e.g., a CLI agent), share whatever you have.

Design answer (½ page max). A data scientist complains: "for some reason my expressions always evaluates to 0 but it does not make any sense, what's going on?" Put on a product manager hat. What feature would you propose to help them? Don't build it — just write the proposal.